

Lista III

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Inicialização

```
library(DBI)
library(RSQLite)

con <- dbConnect(RSQLite::SQLite(), "../datasets/me315.sqlite")
```

Respostas aula

1.

```
SELECT FEE FROM PRESERVE
```

Table 1: Displaying records 1 - 10

FEE
3
0
0
0
0
0
0
0
0
0

Sim. Especialmente é visível diversos valores 0 na tabela apresentada.

2.

```
SELECT ACRES FROM PRESERVE
```

Table 2: Displaying records 1 - 10

ACRES
660
680
49120
121
66
40
830
4
730
1130

Sim. O número de acres não é algo único e pode ser repetido.

3.

```
SELECT DISTINCT FEE FROM PRESERVE
```

Table 3: 2 records

FEE
3
0

```
SELECT DISTINCT ACRES FROM PRESERVE
```

Table 4: Displaying records 1 - 10

ACRES
660

ACRES
680
49120
121
66
40
830
4
730
1130

4.

```
SELECT DISTINCT FEE, ACRES FROM PRESERVE
```

Table 5: Displaying records 1 - 10

FEE	ACRES
3	660
0	680
0	49120
0	121
0	66
0	40
0	830
0	4
0	730
0	1130

5.

```
SELECT PNO FROM PRESERVE
```

Table 6: Displaying records 1 - 10

<u>PNO</u>
1
2
3
5
6
7
9
10
11
12

```
SELECT DISTINCT PNO FROM PRESERVE
```

Table 7: Displaying records 1 - 10

<u>PNO</u>
1
2
3
5
6
7
9
10
11
12

Não há diferença, pois o número da área de preservação é uma chave única.

6.

```
SELECT * FROM PRESERVE WHERE STATE = 'AZ' AND FEE = 0
```

Table 8: 1 records

PNO	PNAME	STATE	ACRES	FEE
7	MULESHOE RANCH	AZ	49120	0

7.

```
SELECT * FROM PRESERVE WHERE STATE = 'AZ' OR FEE = 0
```

Table 9: Displaying records 1 - 10

PNO	PNAME	STATE	ACRES	FEE
5	HASSAYAMPA RIVER	AZ	660	3
3	DANCING PRAIRIE	MT	680	0
7	MULESHOE RANCH	AZ	49120	0
40	SOUTH FORK MADISON	MT	121	0
14	MCELWAIN-OLSEN	MA	66	0
13	TATKON	MA	40	0
9	DAVID H. SMITH	MA	830	0
11	MIACOMET MOORS	MA	4	0
12	MOUNT PLANTAIN	MA	730	0
1	COMERTOWN PRAIRIE	MT	1130	0

8.

```
SELECT PNO, PNAME, ACRES FROM PRESERVE WHERE ACRES > 90 AND ACRES < 1200
```

Table 10: 7 records

PNO	PNAME	ACRES
5	HASSAYAMPA RIVER	660
3	DANCING PRAIRIE	680
40	SOUTH FORK MADISON	121
9	DAVID H. SMITH	830
12	MOUNT PLANTAIN	730
1	COMERTOWN PRAIRIE	1130

PNO	PNAME	ACRES
80	RAMSEY CANYON	380

9.

```
SELECT PNO, PNAME, FEE, ACRES
FROM PRESERVE
WHERE STATE = 'AZ' AND NOT FEE = 0 AND ACRES >= 660 AND ACRES <= 1200
```

Table 11: 2 records

PNO	PNAME	FEE	ACRES
5	HASSAYAMPA RIVER	3	660
6	PAPAGONIA-SONOITA CREEK	3	1200

10

```
SELECT PNO, PNAME, STATE
FROM PRESERVE
WHERE STATE = 'AZ' OR STATE = 'MT'
```

Table 12: 8 records

PNO	PNAME	STATE
5	HASSAYAMPA RIVER	AZ
3	DANCING PRAIRIE	MT
7	MULESHOE RANCH	AZ
40	SOUTH FORK MADISON	MT
1	COMERTOWN PRAIRIE	MT
2	PINE BUTTE SWAMP	MT
80	RAMSEY CANYON	AZ
6	PAPAGONIA-SONOITA CREEK	AZ

11.

```
SELECT PNO, PNAME
FROM PRESERVE
WHERE PNO IN (3, 4, 7, 12, 40)
```

Table 13: 4 records

PNO	PNAME
3	DANCING PRAIRIE
7	MULESHOE RANCH
12	MOUNT PLANTAIN
40	SOUTH FORK MADISON

12.

```
SELECT PNAME, STATE
FROM PRESERVE
WHERE NOT STATE = 'MA'
```

Table 14: 8 records

PNAME	STATE
HASSAYAMPA RIVER	AZ
DANCING PRAIRIE	MT
MULESHOE RANCH	AZ
SOUTH FORK MADISON	MT
COMERTOWN PRAIRIE	MT
PINE BUTTE SWAMP	MT
RAMSEY CANYON	AZ
PAPAGONIA-SONOITA CREEK	AZ

13.

```
SELECT PNAME, STATE, FEE, ACRES
FROM PRESERVE
WHERE (ACRES > 1000 OR STATE = 'AZ') AND FEE = 3
```

Table 15: 3 records

PNAME	STATE	FEE	ACRES
HASSAYAMPA RIVER	AZ	3	660
RAMSEY CANYON	AZ	3	380
PAPAGONIA-SONOITA CREEK	AZ	3	1200

14.

```
SELECT PNAME, STATE, FEE, ACRES
FROM PRESERVE
WHERE ACRES > 1000 OR (STATE = 'AZ' AND FEE = 3)
```

Table 16: 6 records

PNAME	STATE	FEE	ACRES
HASSAYAMPA RIVER	AZ	3	660
MULESHOE RANCH	AZ	0	49120
COMERTOWN PRAIRIE	MT	0	1130
PINE BUTTE SWAMP	MT	0	15000
RAMSEY CANYON	AZ	3	380
PAPAGONIA-SONOITA CREEK	AZ	3	1200

15.

```
SELECT PNO, STATE, FEE
FROM PRESERVE
WHERE NOT (STATE = 'AZ' AND FEE = 3)
```

Table 17: Displaying records 1 - 10

PNO	STATE	FEE
3	MT	0
7	AZ	0
40	MT	0
14	MA	0

PNO	STATE	FEE
13	MA	0
9	MA	0
11	MA	0
12	MA	0
1	MT	0
2	MT	0

16.

```
SELECT PNAME, STATE, FEE, ACRES
FROM PRESERVE
WHERE ACRES < 50 OR (FEE = 3 AND NOT STATE = 'MA')
```

Table 18: 5 records

PNAME	STATE	FEE	ACRES
HASSAYAMPA RIVER	AZ	3	660
TATKON	MA	0	40
MIACOMET MOORS	MA	0	4
RAMSEY CANYON	AZ	3	380
PAPAGONIA-SONOITA CREEK	AZ	3	1200

Respostas hands-on

1.

```
SELECT DISTINCT STATE, FEE
FROM PRESERVE
```

Table 19: 4 records

STATE	FEE
AZ	3
MT	0
AZ	0

STATE	FEE
MA	0

2.

```
SELECT DISTINCT DNO
FROM EMPLOYEE
```

Table 20: 3 records

DNO
20
10
40

3.

```
SELECT DNO, SALARY FROM EMPLOYEE
```

Table 21: 6 records

DNO	SALARY
20	2000
10	2000
20	3000
40	500
10	400
20	9000

```
SELECT DISTINCT DNO, SALARY FROM EMPLOYEE
```

Table 22: 6 records

DNO	SALARY
20	2000
10	2000
20	3000
40	500
10	400
20	9000

```
SELECT DISTINCT DNO, SALARY FROM EMPLOYEE ORDER BY DNO, SALARY
```

Table 23: 6 records

DNO	SALARY
10	400
10	2000
20	2000
20	3000
20	9000
40	500

Como tanto com ou sem DISTINCT o número de linhas é 6, é possível assumir que não há salários repetidos nos departamentos. A última tabela está ordenada.

4.

```
SELECT *
FROM PRESERVE
WHERE STATE = 'MT' AND ACRES < 1000
```

Table 24: 2 records

PNO	PNAME	STATE	ACRES	FEE
3	DANCING PRAIRIE	MT	680	0
40	SOUTH FORK MADISON	MT	121	0

5.

```
SELECT *  
FROM PRESERVE  
WHERE ACRES >= 1200 AND ACRES <= 15000
```

Table 25: 2 records

PNO	PNAME	STATE	ACRES	FEE
2	PINE BUTTE SWAMP	MT	15000	0
6	PAPAGONIA-SONOITA CREEK	AZ	1200	3

6.

```
SELECT *  
FROM PRESERVE  
WHERE STATE = 'MT' AND FEE = 0 AND ACRES < 10000
```

Table 26: 3 records

PNO	PNAME	STATE	ACRES	FEE
3	DANCING PRAIRIE	MT	680	0
40	SOUTH FORK MADISON	MT	121	0
1	COMERTOWN PRAIRIE	MT	1130	0

7.

```
SELECT *  
FROM PRESERVE  
WHERE STATE IN ('MT', 'MA')
```

Table 27: Displaying records 1 - 10

PNO	PNAME	STATE	ACRES	FEE
3	DANCING PRAIRIE	MT	680	0
40	SOUTH FORK MADISON	MT	121	0
14	MCELWAIN-OLSEN	MA	66	0
13	TATKON	MA	40	0
9	DAVID H. SMITH	MA	830	0
11	MIACOMET MOORS	MA	4	0
12	MOUNT PLANTAIN	MA	730	0
1	COMERTOWN PRAIRIE	MT	1130	0
2	PINE BUTTE SWAMP	MT	15000	0
10	HOFT FARM	MA	90	0

8.

```
SELECT *
FROM PRESERVE
WHERE STATE = 'MT' OR ACRES < 1000
```

Table 28: Displaying records 1 - 10

PNO	PNAME	STATE	ACRES	FEE
5	HASSAYAMPA RIVER	AZ	660	3
3	DANCING PRAIRIE	MT	680	0
40	SOUTH FORK MADISON	MT	121	0
14	MCELWAIN-OLSEN	MA	66	0
13	TATKON	MA	40	0
9	DAVID H. SMITH	MA	830	0
11	MIACOMET MOORS	MA	4	0
12	MOUNT PLANTAIN	MA	730	0
1	COMERTOWN PRAIRIE	MT	1130	0
2	PINE BUTTE SWAMP	MT	15000	0

9.

```
SELECT PNO, PNAME
FROM PRESERVE
WHERE NOT FEE = 0
```

Table 29: 3 records

PNO	PNAME
5	HASSAYAMPA RIVER
80	RAMSEY CANYON
6	PAPAGONIA-SONOITA CREEK

10.

```
SELECT PNO, PNAME
FROM PRESERVE
WHERE NOT FEE = 3 AND NOT FEE = 10
```

Table 30: Displaying records 1 - 10

PNO	PNAME
3	DANCING PRAIRIE
7	MULESHOE RANCH
40	SOUTH FORK MADISON
14	MCELWAIN-OLSEN
13	TATKON
9	DAVID H. SMITH
11	MIACOMET MOORS
12	MOUNT PLANTAIN
1	COMERTOWN PRAIRIE
2	PINE BUTTE SWAMP

11.

```
SELECT *
FROM PRESERVE
WHERE STATE = 'AZ' AND FEE = 0 OR ACRES < 100
```

Table 31: 5 records

PNO	PNAME	STATE	ACRES	FEE
7	MULESHOE RANCH	AZ	49120	0
14	MCELWAIN-OLSEN	MA	66	0
13	TATKON	MA	40	0
11	MIACOMET MOORS	MA	4	0
10	HOFT FARM	MA	90	0

12.

```
SELECT *
FROM PRESERVE
WHERE ACRES < 1000 AND FEE = 0 OR STATE = 'AZ'
```

Table 32: Displaying records 1 - 10

PNO	PNAME	STATE	ACRES	FEE
5	HASSAYAMPA RIVER	AZ	660	3
3	DANCING PRAIRIE	MT	680	0
7	MULESHOE RANCH	AZ	49120	0
40	SOUTH FORK MADISON	MT	121	0
14	MCELWAIN-OLSEN	MA	66	0
13	TATKON	MA	40	0
9	DAVID H. SMITH	MA	830	0
11	MIACOMET MOORS	MA	4	0
12	MOUNT PLANTAIN	MA	730	0
80	RAMSEY CANYON	AZ	380	3

13.

```
SELECT *
FROM PRESERVE
WHERE FEE > 0 OR (STATE = 'MT' AND ACRES > 1000)
```

Table 33: 5 records

PNO	PNAME	STATE	ACRES	FEE
5	HASSAYAMPA RIVER	AZ	660	3
1	COMERTOWN PRAIRIE	MT	1130	0
2	PINE BUTTE SWAMP	MT	15000	0
80	RAMSEY CANYON	AZ	380	3
6	PAPAGONIA-SONOITA CREEK	AZ	1200	3

14.

```
SELECT *
FROM PRESERVE
WHERE NOT (STATE = 'MT' AND FEE = 0)
```

Table 34: Displaying records 1 - 10

PNO	PNAME	STATE	ACRES	FEE
5	HASSAYAMPA RIVER	AZ	660	3
7	MULESHOE RANCH	AZ	49120	0
14	MCELWAIN-OLSEN	MA	66	0
13	TATKON	MA	40	0
9	DAVID H. SMITH	MA	830	0
11	MIACOMET MOORS	MA	4	0
12	MOUNT PLANTAIN	MA	730	0
80	RAMSEY CANYON	AZ	380	3
10	HOFT FARM	MA	90	0
6	PAPAGONIA-SONOITA CREEK	AZ	1200	3

15.

Sim, pois o operador AND é distributível sobre o OR.

16.

Sim, também é distributível pela ordem contrária.

17.

Não, a negação de um **AND** é o **OR** das negações, não **AND** das negações como segunda linha faz. Primeira e terceira linhas, são iguais, mas diferentes da segunda.

18.

Não, mesmo caso do item 17. Primeira e terceira linhas, são iguais, mas diferentes da segunda.